

INFS 343-Lab 2:

Part I: Let's try some basic R commands here. Remember to include the screenshot for your output.

1.

```
#This is the instruction
#Introduce yourself to me
#Change the code to reflect your information

name <- "Ning"
major <- "MIS"
age <- 21
hobby <- "cooking"

#Why we need to write down all the variables here?
name
major
age
hobby
,
```

Make sure that you change the information here accordingly!! Your output:

```
> name
[1] "Marcelo Andrada Batista"
> major1
[1] "Information Systems"
> major2
[1] "Marketing"
> age1
[1] 21
> hobby1
[1] "clubbing"
```

2.

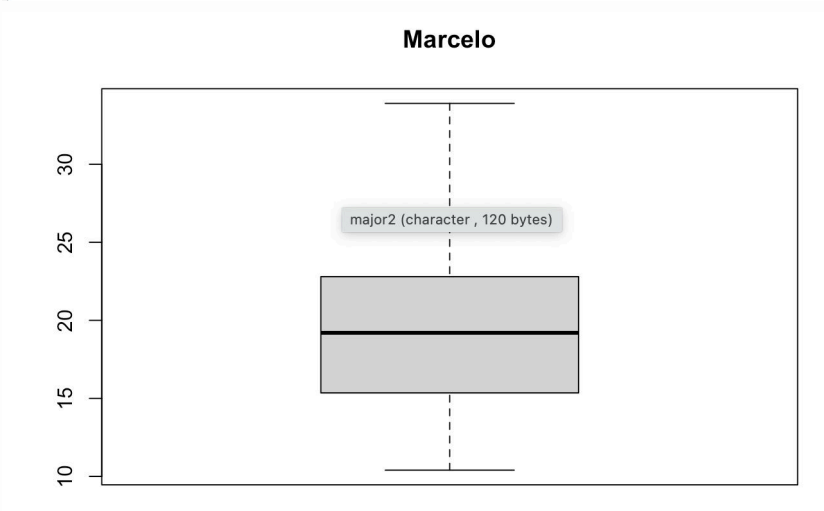
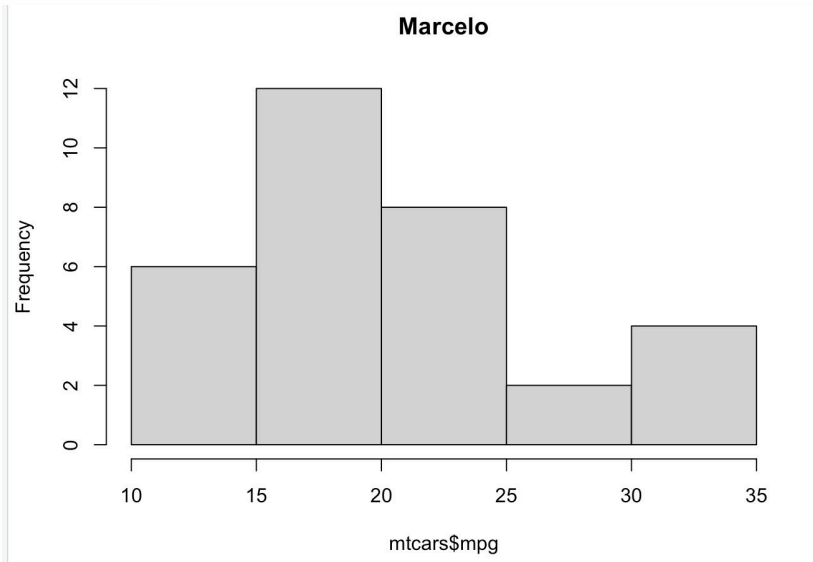
```
#histogram, change the title to your name
hist(mtcars$mpg, main= "Ning Yang")

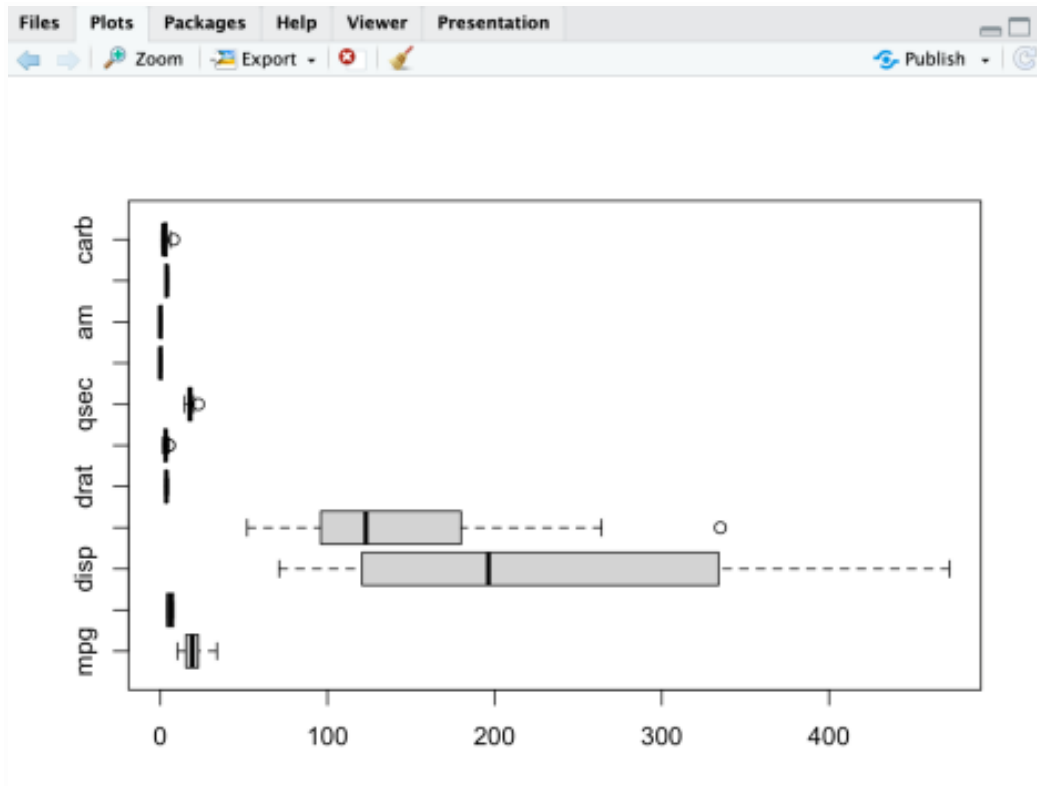
#boxplot, change the title to your name
boxplot(mtcars$mpg, main = "Ning Yang")

|
#what do you see when you run boxplot(mtcars)?
boxplot(mtcars, horizontal= TRUE)
```

Make sure that you change the title into your own name!! What does horizontal = TRUE do? Try it with and without horizontal = True to find out!

Your output:





```
print(mtcars)
summary(mtcars)
```

1. Please screenshot your output here.

```
> #try without horizontal = true
> print(mtcars)
```

	mpg	cyl	displacement	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4
Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4
Merc 450SE	16.4	8	275.8	180	3.07	4.070	17.40	0	0	3	3
Merc 450SL	17.3	8	275.8	180	3.07	3.730	17.60	0	0	3	3
Merc 450SLC	15.2	8	275.8	180	3.07	3.780	18.00	0	0	3	3
Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.250	17.98	0	0	3	4
Lincoln Continental	10.4	8	460.0	215	3.00	5.424	17.82	0	0	3	4
Chrysler Imperial	14.7	8	440.0	230	3.23	5.345	17.42	0	0	3	4
Fiat 128	32.4	4	78.7	66	4.08	2.200	19.47	1	1	4	1
Honda Civic	30.4	4	75.7	52	4.93	1.615	18.52	1	1	4	2
Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.90	1	1	4	1
Toyota Corona	21.5	4	120.1	97	3.70	2.465	20.01	1	0	3	1
Dodge Challenger	15.5	8	318.0	150	2.76	3.520	16.87	0	0	3	2
AMC Javelin	15.2	8	304.0	150	3.15	3.435	17.30	0	0	3	2
Camaro Z28	13.3	8	350.0	245	3.73	3.840	15.41	0	0	3	4
Pontiac Firebird	19.2	8	400.0	175	3.08	3.845	17.05	0	0	3	2
Fiat X1-9	27.3	4	79.0	66	4.08	1.935	18.90	1	1	4	1
Porsche 914-2	26.0	4	120.3	91	4.43	2.140	16.70	0	1	5	2
Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.90	1	1	5	2
Ford Pantera L	15.8	8	351.0	264	4.22	3.170	14.50	0	1	5	4
Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.50	0	1	5	6
Maserati Bora	15.0	8	301.0	335	3.54	3.570	14.60	0	1	5	8
Volvo 142E	21.4	4	121.0	109	4.11	2.780	18.60	1	1	4	2

```
> summary(mtcars)
```

mpg	cyl	disp	hp	drat	wt
Min. :10.40	Min. :4.000	Min. : 71.1	Min. : 52.0	Min. :2.760	Min. :1.513
1st Qu.:15.43	1st Qu.:4.000	1st Qu.:120.8	1st Qu.: 96.5	1st Qu.:3.080	1st Qu.:2.581
Median :19.20	Median :6.000	Median :196.3	Median :123.0	Median :3.695	Median :3.325
Mean :20.09	Mean :6.188	Mean :230.7	Mean :146.7	Mean :3.597	Mean :3.217
3rd Qu.:22.80	3rd Qu.:8.000	3rd Qu.:326.0	3rd Qu.:180.0	3rd Qu.:3.920	3rd Qu.:3.610
Max. :33.90	Max. :8.000	Max. :472.0	Max. :335.0	Max. :4.930	Max. :5.424

qsec	vs	am	gear	carb
Min. :14.50	Min. :0.0000	Min. :0.0000	Min. :3.000	Min. :1.000
1st Qu.:16.89	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:3.000	1st Qu.:2.000
Median :17.71	Median :0.0000	Median :0.0000	Median :4.000	Median :2.000
Mean :17.85	Mean :0.4375	Mean :0.4062	Mean :3.688	Mean :2.812
3rd Qu.:18.90	3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:4.000	3rd Qu.:4.000
Max. :22.90	Max. :1.0000	Max. :1.0000	Max. :5.000	Max. :8.000

2. Based on the output, how many variables are in mtcars? Please list all of them.

11 (mpg, cyl, disp, hp, drat, wt, qsec, vs, am, gear, carb)

3. In the statistical summary, what are the types of statistics included here for each variable?

Minimum, Q1, Median, Mean, Q3, Max

Part III: Write a R program to get the structure of a given data frame. Please make sure you type the data frame name as homework_data.

```
homework_data = data.frame(
  name = c('Ava', 'Arthur', 'Conrad', 'Kathryn', 'Kaia', 'Jack',
    'Mallory', 'Riley', 'Ryan', 'Zoe'),
  score = c(3, 4, 3.5, 2, 3.5, 0.5, 2.5, 3.5, 4, 4),
  attempts = c(1, 3, 2, 3, 2, 3, 1, 1, 2, 1),
  qualify = c('yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no',
    'yes')
)
print("Original dataframe:")
print(homework_data)
print("Structure of the said data frame:")
print(str(homework_data))
```

1. Please screenshot your output here.

```

>
> homework_data = data.frame()
> name = c('Ava', 'Arthur', 'Conrad', 'Kathryn', 'Kaia', 'Jack', 'Mallory', 'Riley', 'Ryan', 'Zoe')
> score = c(3,4,3.5,2,3.5,0.5,2.5,3.5,4,4),

```

Error: unexpected ',' in "score = c(3,4,3.5,2,3.5,0.5,2.5,3.5,4,4),"

Show Traceback
Rerun with Debug

```

> score = c(3,4,3.5,2,3.5,0.5,2.5,3.5,4,4)
> attempts = c(1,3,2,3,2,3,1,1,2,1)
> qualify = c('yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes')
>
> print("Original dataframe:")
[1] "Original dataframe:"
> print(homework_data)
data frame with 0 columns and 0 rows
> homework_data=data.frame(name,score,attempts,qualify)
> print(homework_data)
  name score attempts qualify
1   Ava   3.0         1    yes
2 Arthur   4.0         3     no
3 Conrad   3.5         2    yes
4 Kathryn   2.0         3     no
5   Kaia   3.5         2     no
6   Jack   0.5         3    yes
7 Mallory   2.5         1    yes
8   Riley   3.5         1     no
9    Ryan   4.0         2     no
10   Zoe   4.0         1    yes

```

2. How to check whether there is missing value in the dataset?

```

> colSums(is.na(homework_data))
  name      score attempts  qualify
    0         0         0         0

```

3. How to sort the students by their grade from the highest to the lowest?

```
> homework_data[order(homework_data$score)]
```

```
Error in `[.data.frame'`(homework_data, order(homework_data$score)) :  
  undefined columns selected
```

```
> homework_data[order(-homework_data$score)]
```

```
Error in `[.data.frame'`(homework_data, order(-homework_data$score)) :  
  undefined columns selected
```

```
> homework_data[order(homework_data$score),]
```

	name	score	attempts	qualify
6	Jack	0.5	3	yes
4	Kathryn	2.0	3	no
7	Mallory	2.5	1	yes
1	Ava	3.0	1	yes
3	Conrad	3.5	2	yes
5	Kaia	3.5	2	no
8	Riley	3.5	1	no
2	Arthur	4.0	3	no
9	Ryan	4.0	2	no
10	Zoe	4.0	1	yes

```
>
```

4. How to calculate the mean of the students scores?

```
> mean(homework_data$score)
```

```
[1] 3.05
```

5. Based on the data, who performed the best and why? You need to convince me with numbers.

Arthur, Ryan, and Zoe (Only those 3 got 4.0)

6. Who performed the worst and why? You need to convince me with numbers.

Jack. Only one that scored below a 1

